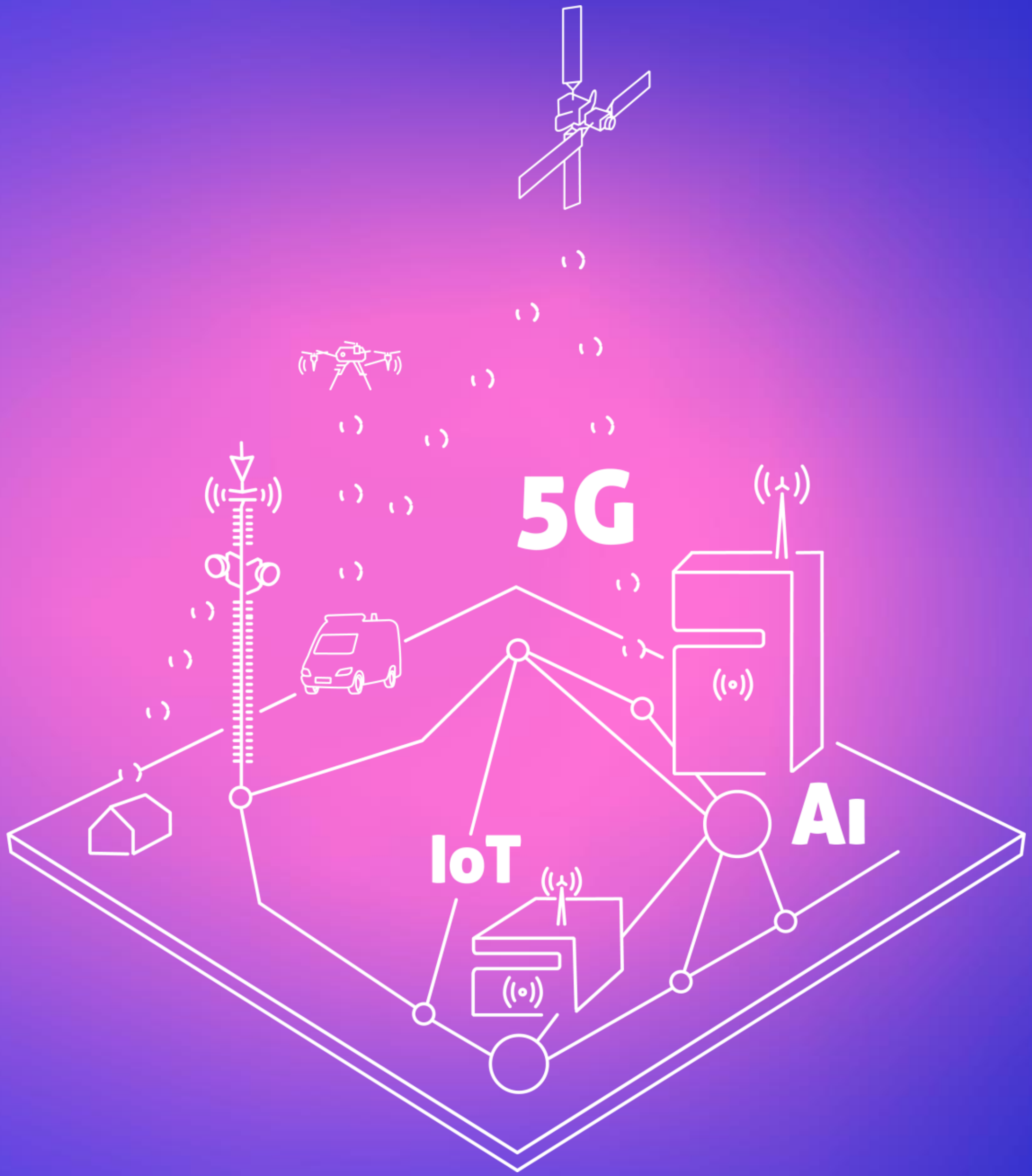
CELLULAR EXPERT

2. Line of Sight (Profile)



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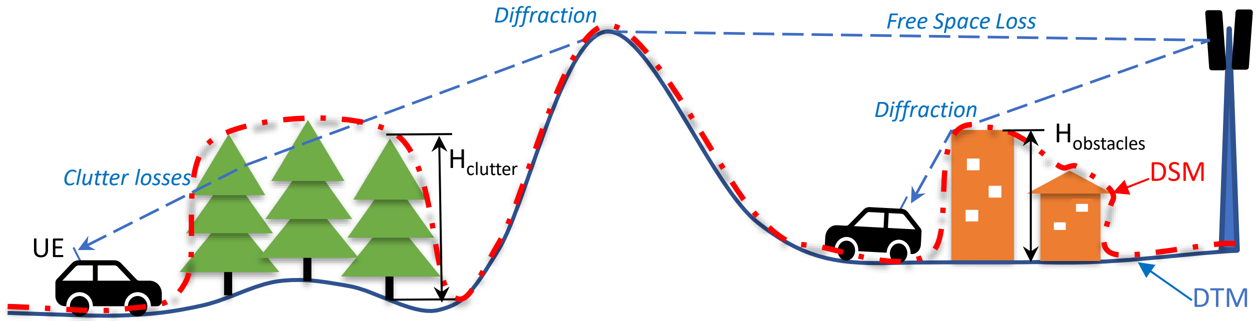
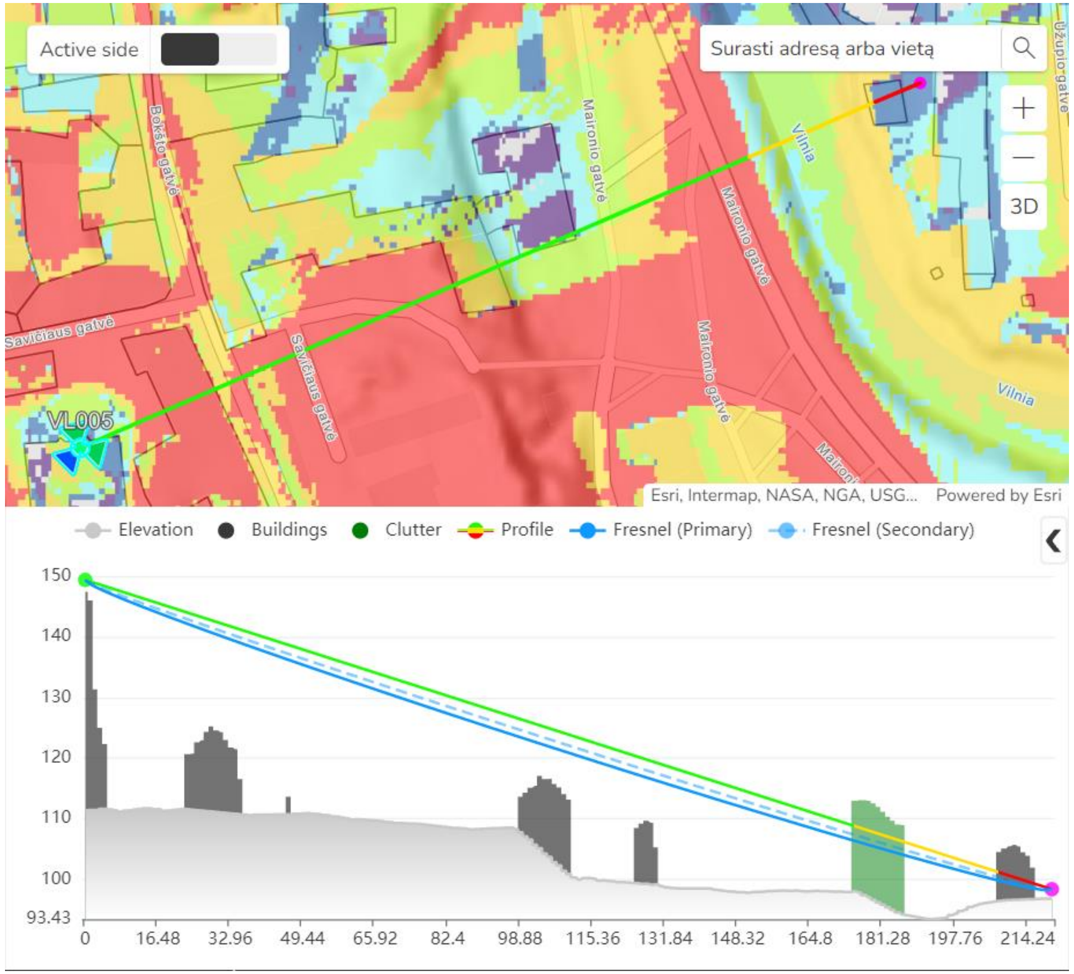
Modelling Outdoor coverage

The CE tools make use of three distinct GIS data layers to obtain high precision modelling of radio wave propagation losses:

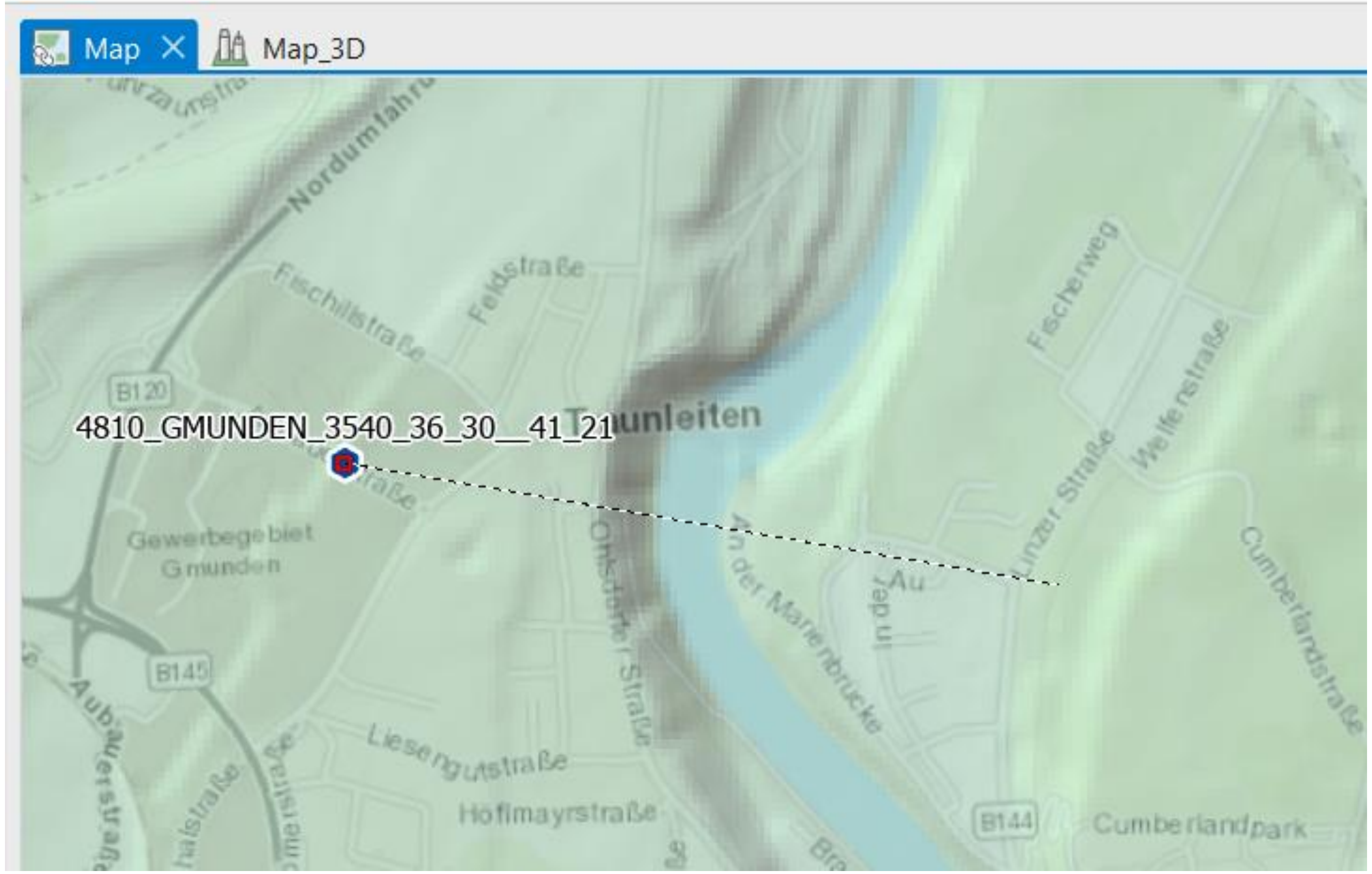
1. **Digital Terrain Model** (DTM), also known as Digital Elevation Model (DEM), which describes Earth surface, i.e., path terrain profile in terms of ground elevation above uniform sea level.
2. **Obstacles layer**, delineating buildings and other such objects above Earth surface that may be considered to be principal impediments for radio wave propagation.
3. **Clutter layer**, delineating natural occurring or human cultivated ground cover that may be partially penetrable by radio waves, such as natural vegetation (e.g., forests, trees, bushes) or various crops, gardens, parks, etc.





Point to point (Profile) input

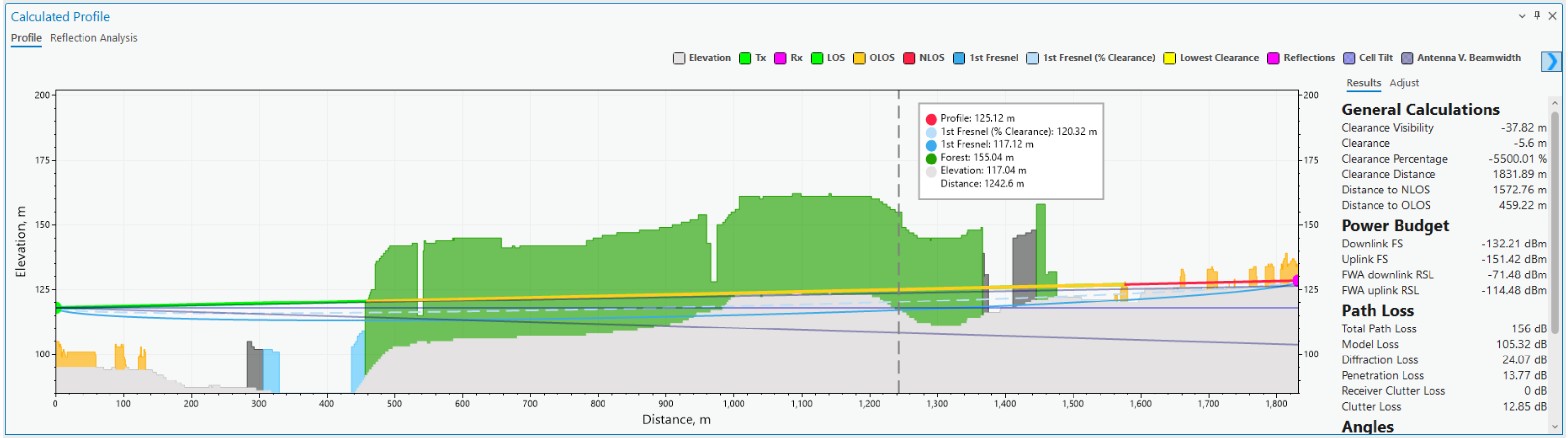
- Geodata
  - Elevation
  - Buildings
  - Clutter
- Frequency
- Fresnel zone (%)
- Earth radius
- Transmitter
  - Height
  - Power
- Receiver
  - Height
  - Power



Profile calculations

CELLULAR EXPERT

- General
  - Clearance
  - Clearance Percentage
  - Clearance Distance
  - Distance to NLOS
  - Distance to OLOS
- Power Budget
  - Downlink FS
  - Uplink FS
  - FWA downlink RSL
  - FWA uplink RSL
- Path Loss
  - Total Path Loss
    - Model Loss
    - Diffraction Loss
    - Penetration Loss
    - Receiver Clutter Loss
    - Clutter Loss
- Angles

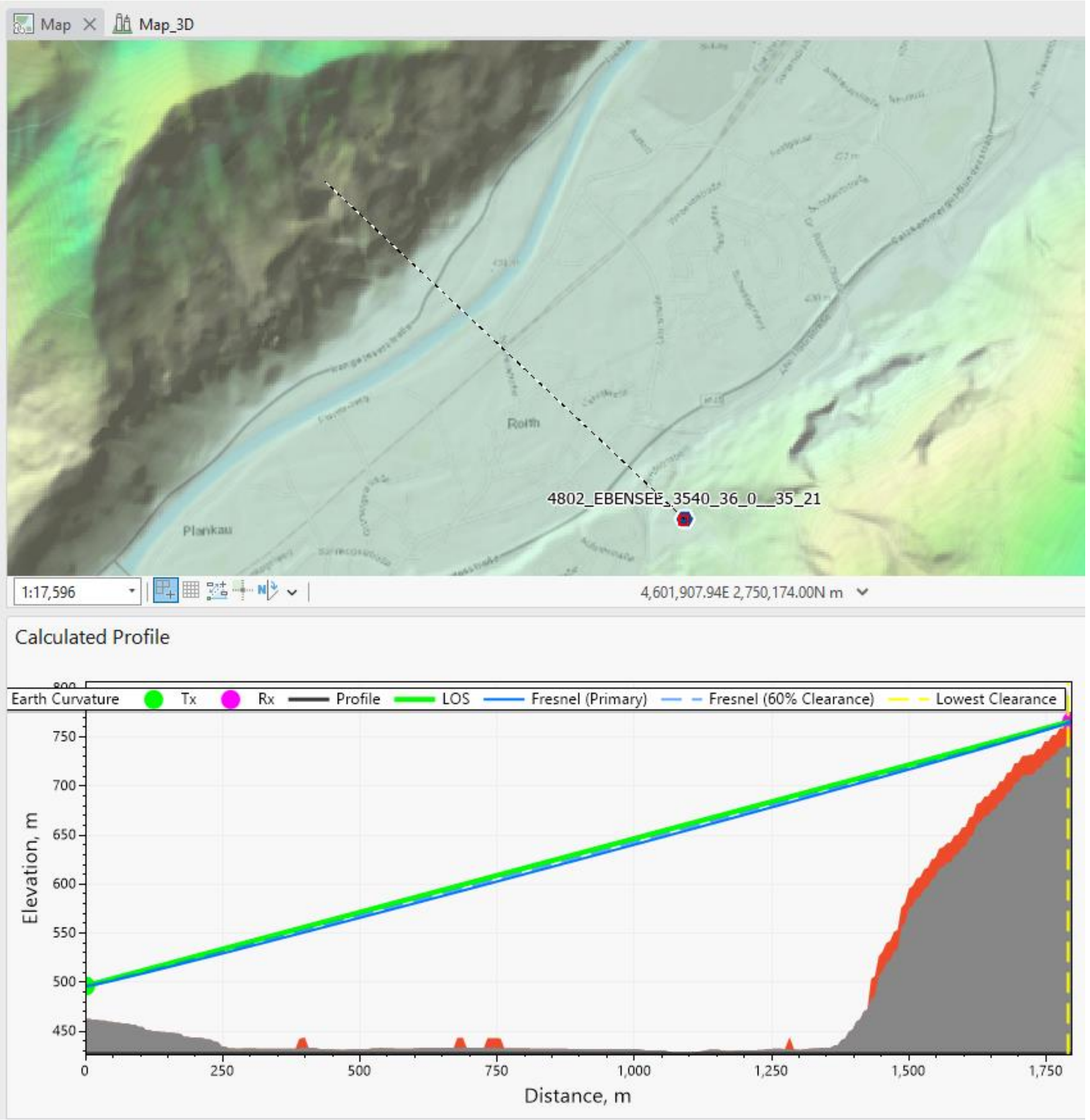


General Calculations	
Clearance Visibility	-37.82 m
Clearance	-5.6 m
Clearance Percentage	-5500.01 %
Clearance Distance	1831.89 m
Distance to NLOS	1572.76 m
Distance to OLOS	459.22 m
Power Budget	
Downlink FS	-132.21 dBm
Uplink FS	-151.42 dBm
FWA downlink RSL	-71.48 dBm
FWA uplink RSL	-114.48 dBm
Path Loss	
Total Path Loss	156 dB
Model Loss	105.32 dB
Diffraction Loss	24.07 dB
Penetration Loss	13.77 dB
Receiver Clutter Loss	0 dB
Clutter Loss	12.85 dB

Angles	
Transmitter azimuth	40 °
Transmitter tilt	0 °
TX-RX azimuth	90 °
TX-RX tilt	-0.33 °
Receiver azimuth	270 °
Receiver tilt	0.33 °
RX-TX azimuth	270 °
RX-TX tilt	0.33 °

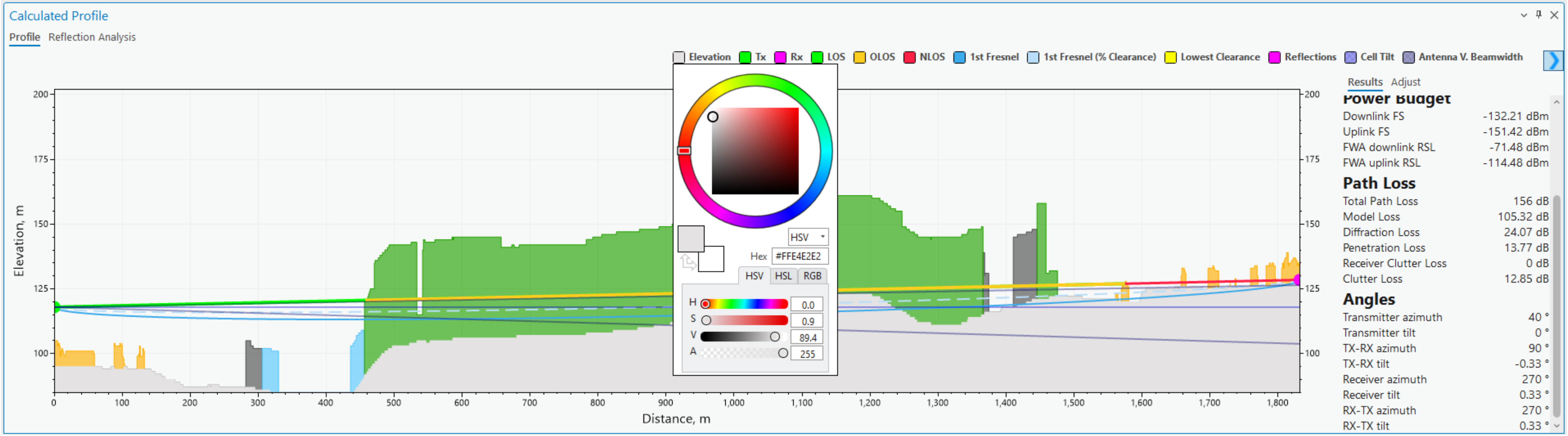
Dynamic profile

- Fix transmitter
- Dynamic option

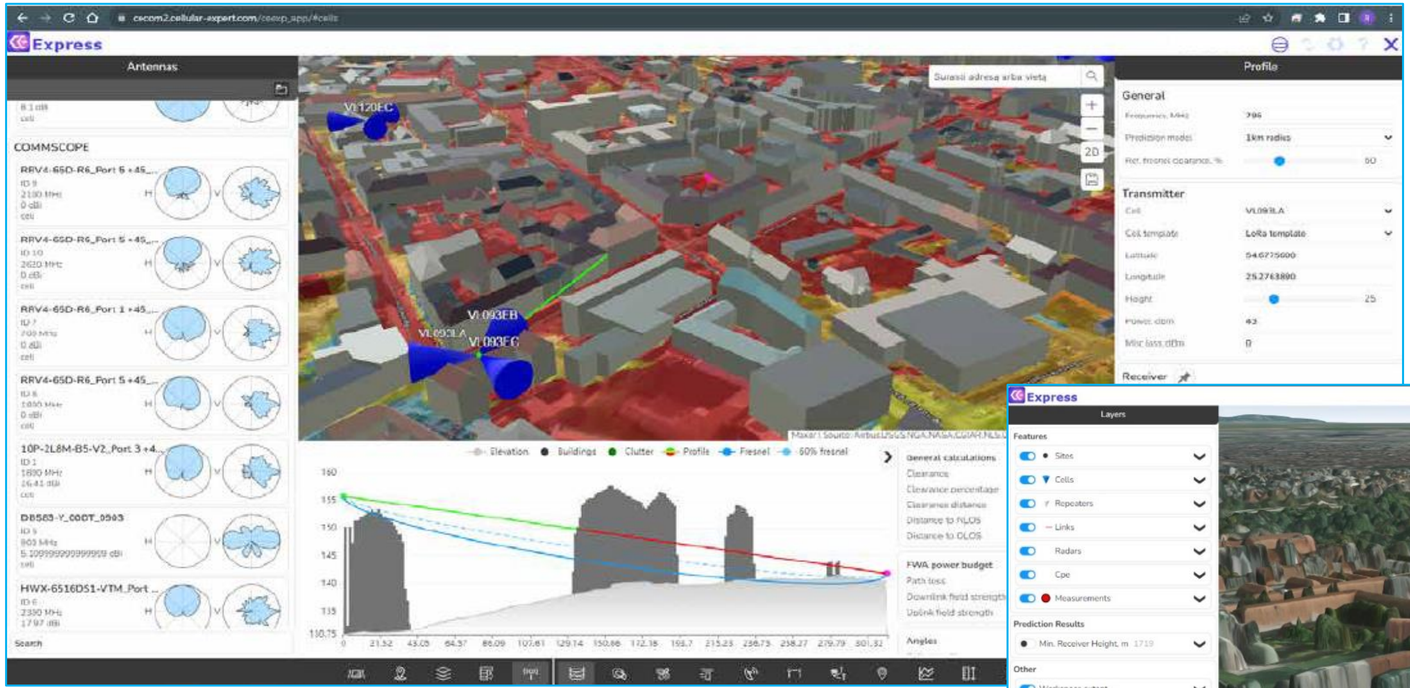


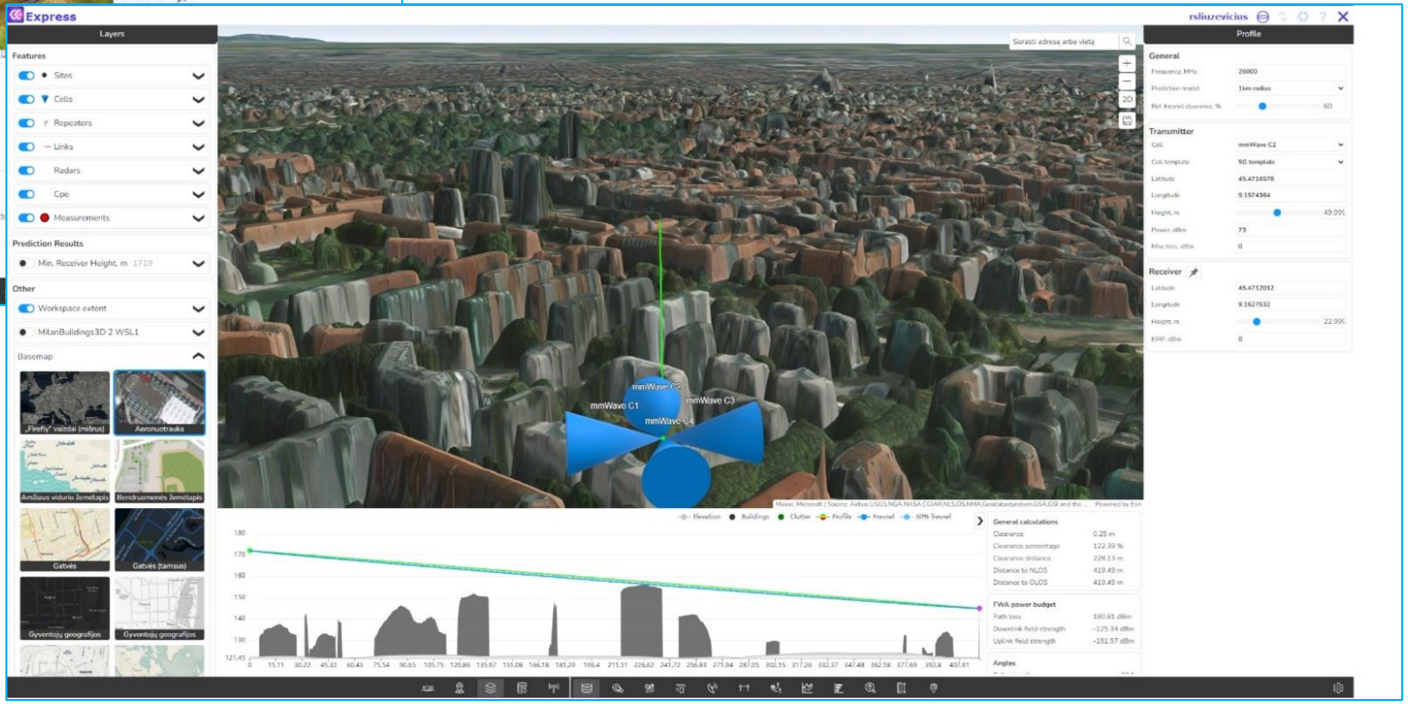
Profile symbology

- Define colors for each object in profile

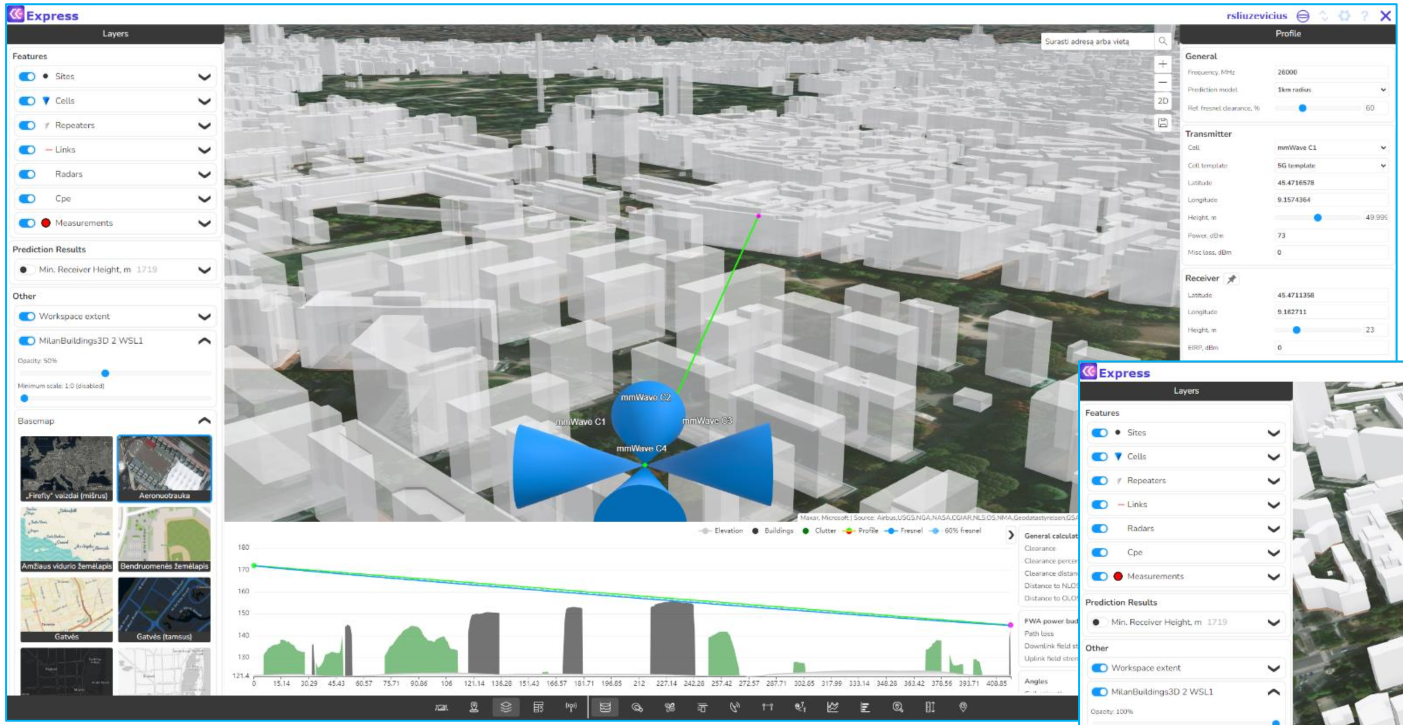


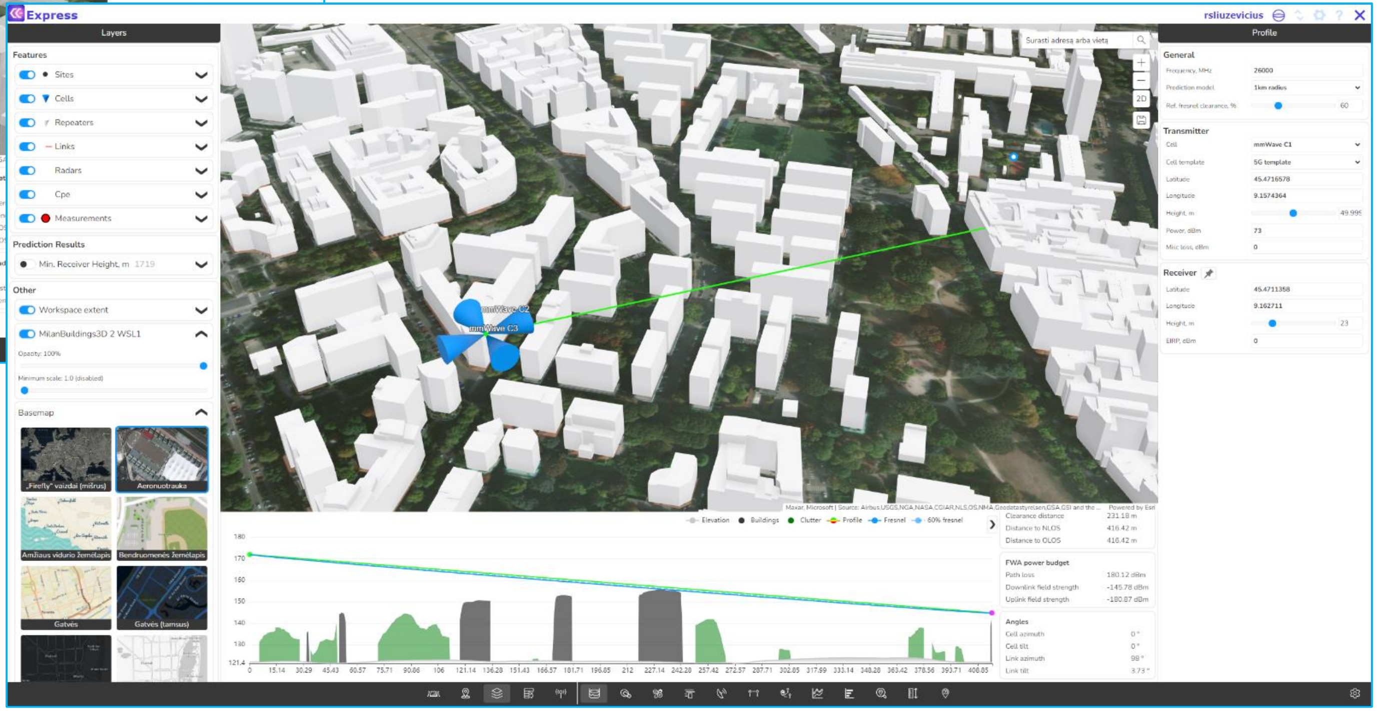
Profile 3D





Profile 3D


Express
rsliuzevicius
Layers
Profile
Features
Sites
Cells
Repeaters
Links
Radars
Cpe
Measurements
Prediction Results
Min. Receiver Height, m
Other
Workspace extent
MilanBuildings3D 2 WSL1
Basemap
„Firefly" vaizdai (mišrus)
Aeronuotrauka
Amžiaus vidurio žemėlapis
Bendruomenės žemėlapis
Gatvės
Gatvės (tamsus)
General
Frequency, MHz
26000
Prediction model
1km radius
Transmitter
Cell
mmWave C1
Cell template
5G template
Latitude
45.4716578
Longitude
9.1574364
Receiver
45.4711358
9.162711
mmWave C1
mmWave C2
mmWave C3
mmWave C4
Elevation
Buildings
Clutter
Profile
Fresnel
60% fresnel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Amžiaus vidurio žemėlapis
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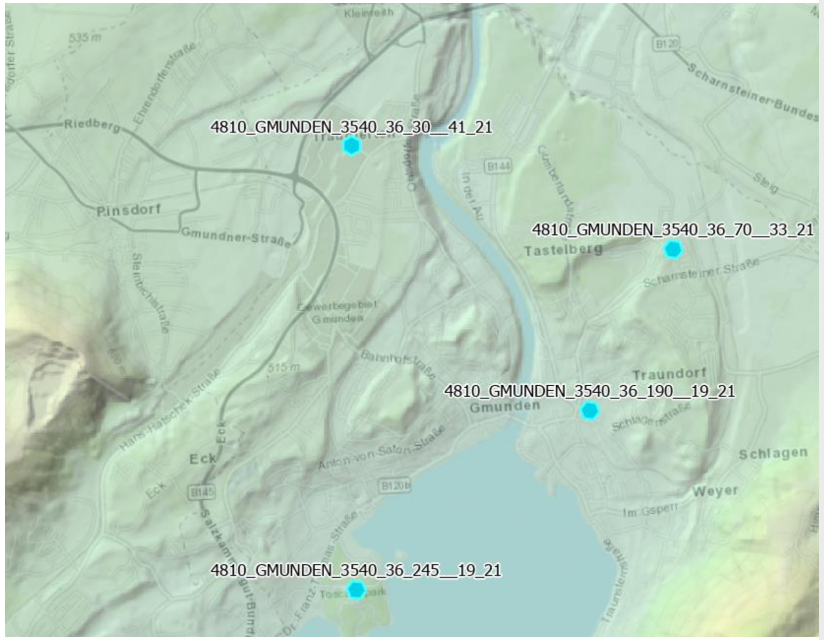
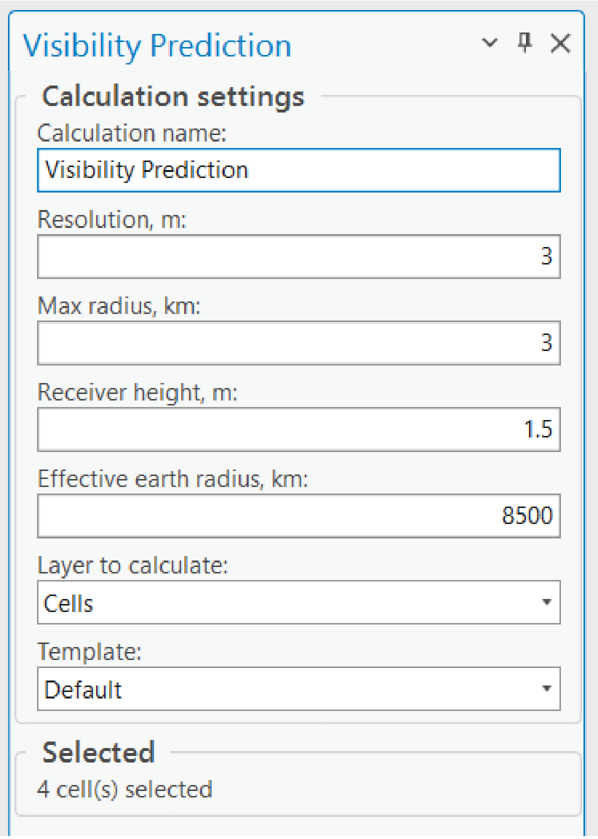
Gatvės
Gatvės (tamsus)
General
Transmitter
Receiver
FWA power budget
Angles

Visibility prediction input

- Geodata
  - Elevation
  - Obstacle
- Frequency
- Calculation radius
- Earth radius
- Transmitter
- Receiver (height)

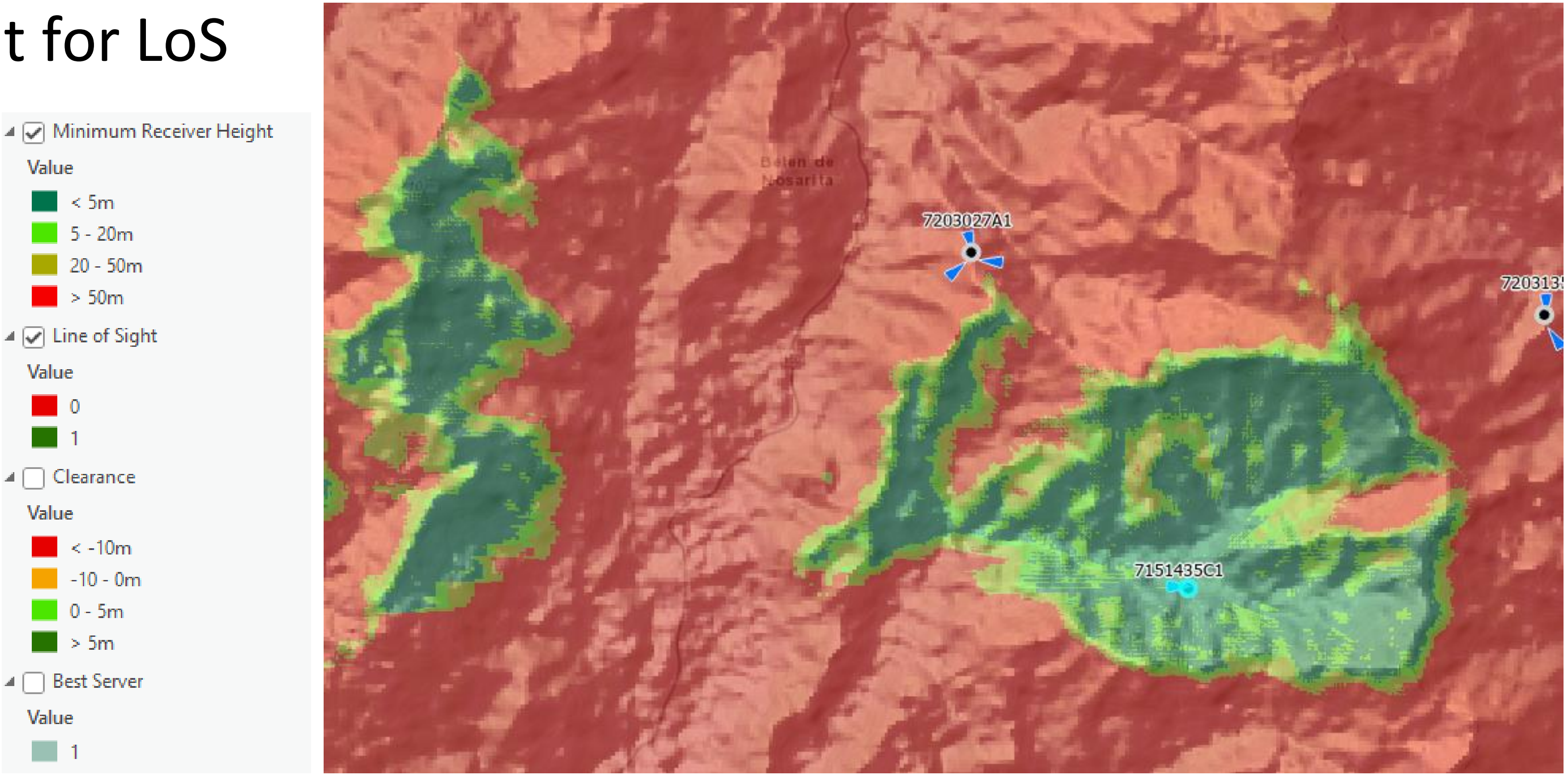




Visibility Results

CELLULAR EXPERT

- Line of Sight
- Required height for LoS
- Clearance



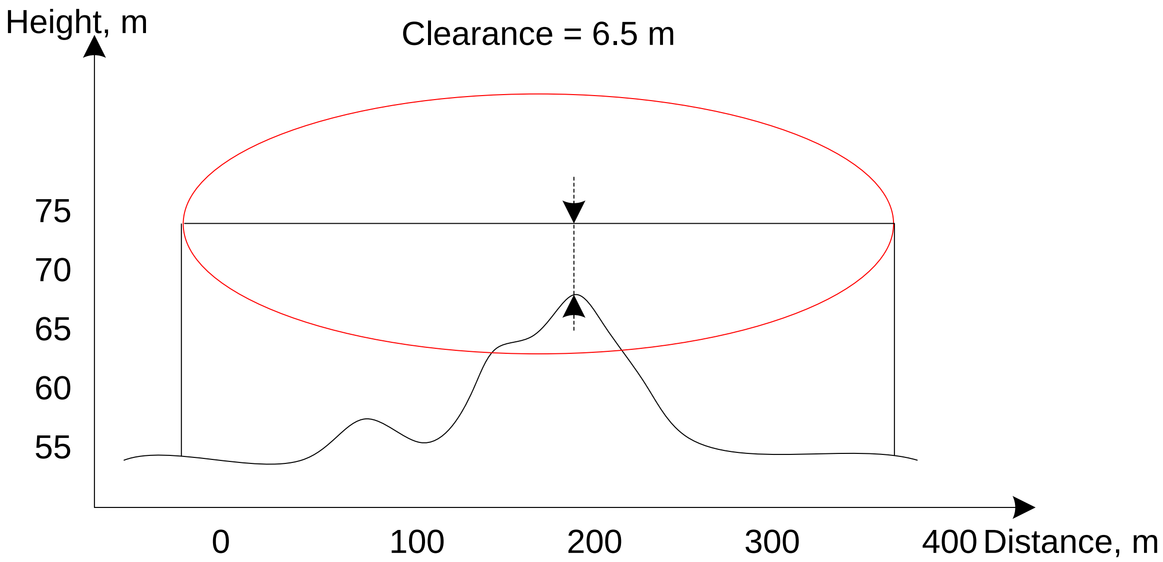
Visibility: Line of Sight

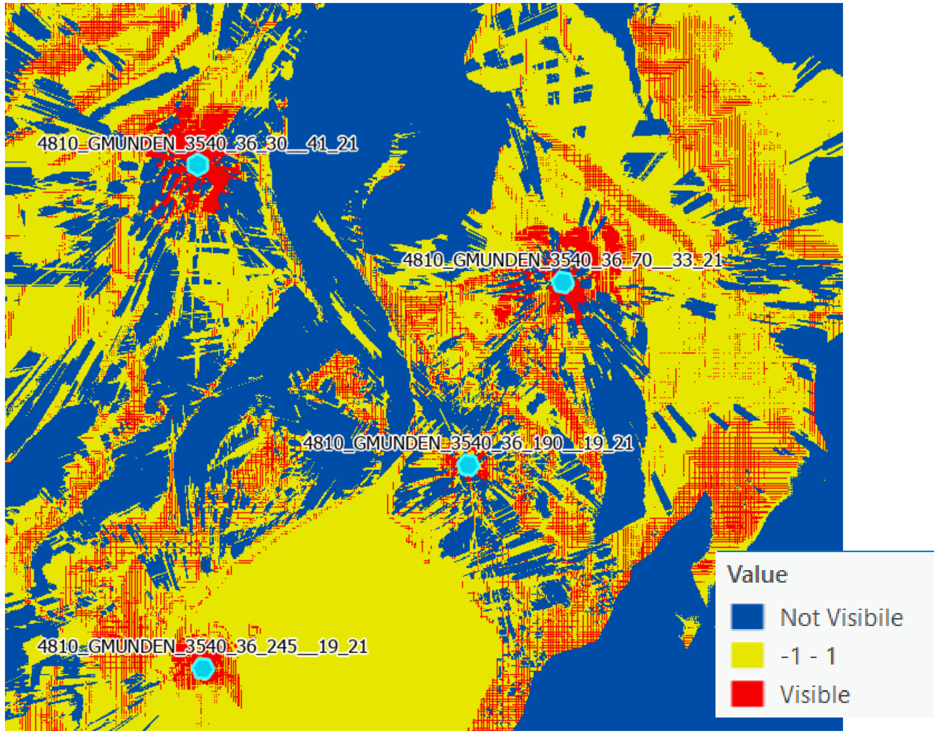
Possible values:

- 0
- 1

Visibility: Clearance

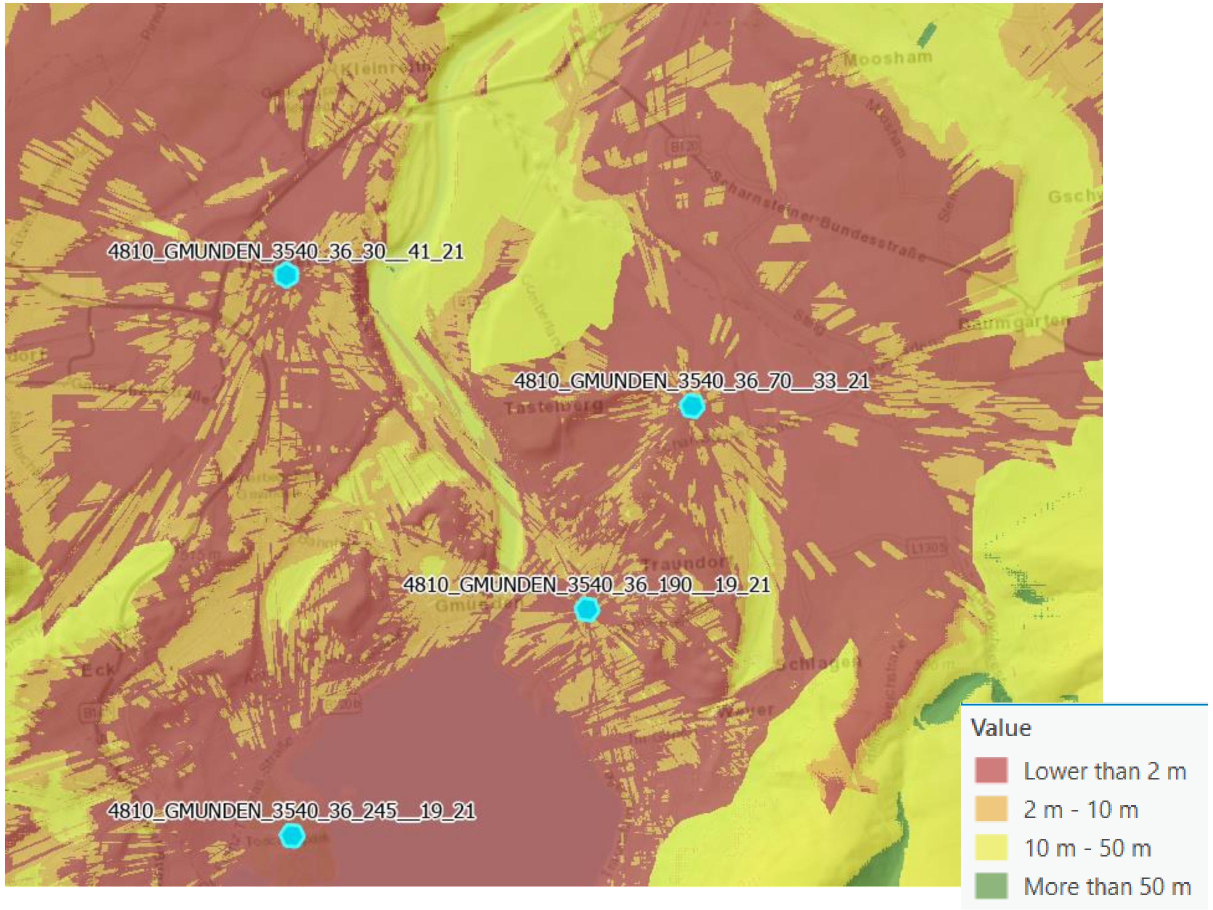
- Clearance – clearance of visibility line between Tx to Rx as a distance in meters



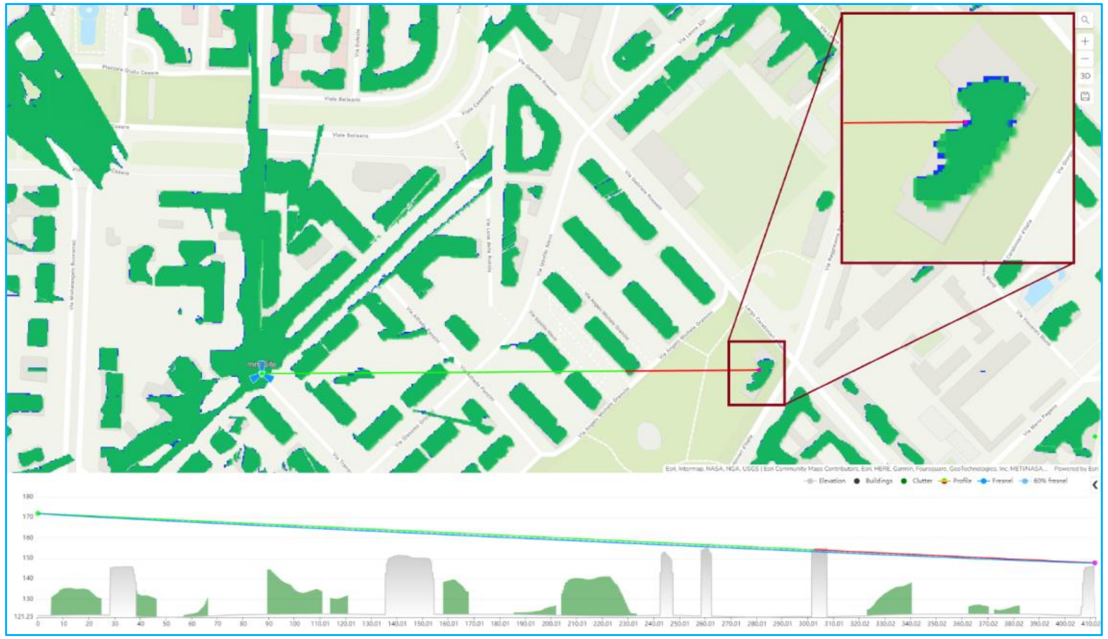


Visibility: Minimum Receiver Height

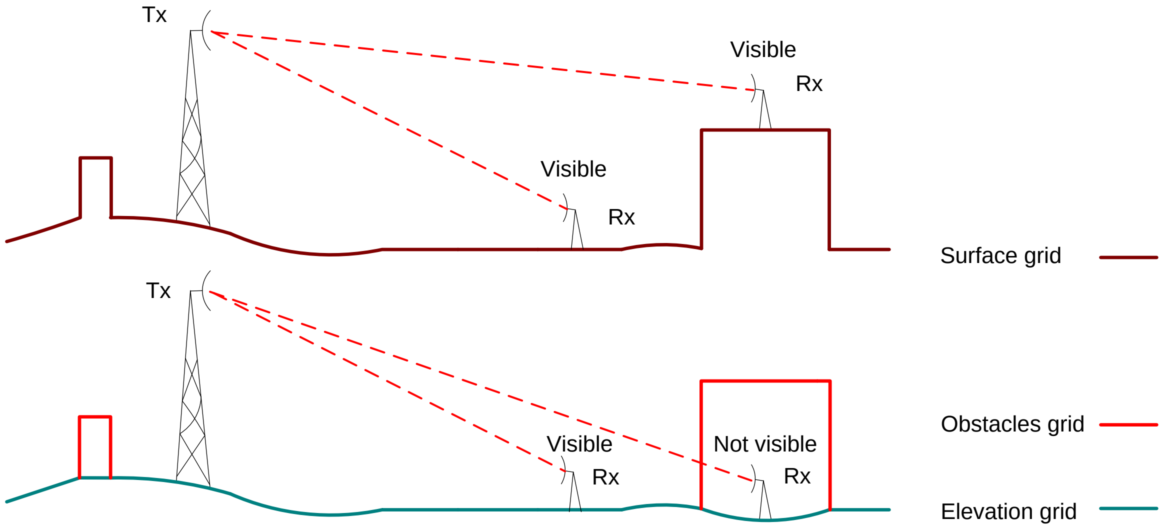
- Receiver height – calculate required receiver heights to have a visibility

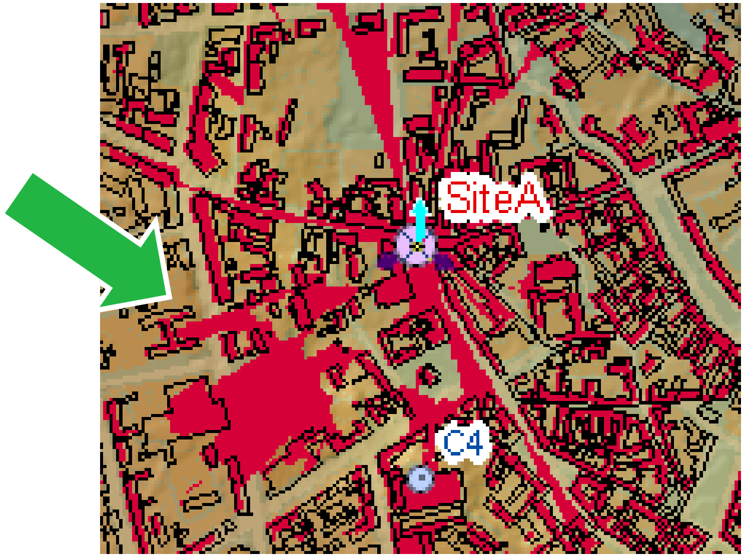


Visibility: Line of Sight sum meter topo data



Surface vs Eelevation+Obstacles





Surface

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Bendruomenės žemėlapis
Profile
General
Transmitter
Receiver
General calculations
FWA power budget
Angles
Profile: 156.83 m
Fresnel: 155.74 m
60% fresnel: 156.18 m
Buildings: 154.97 m
Clutter: 154.97 m
Elevation: 154.97 m
Distance: 234.25 m

DTM + Obstacles (Buildings/Vegetation)

DTM + Buildings + Vegetation

Exercise

Description: C:\CE_Course\0. Descriptions

Name: 2. Line of Sight (Profile).pdf

CELLULAR EXPERT

Thank you!

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